

TCP and UDP

Overview: TCP and UDP are transport-layer protocols used to deliver data between applications running on networked devices. They use port numbers to identify communicating applications, but provide different communication models.

TCP focuses on reliable, ordered delivery and connection management, while UDP focuses on low overhead and connectionless delivery without built-in guarantees for delivery or ordering.

1. TCP — Transmission Control Protocol

TCP is a **connection-oriented** transport protocol designed to provide reliable, ordered delivery of a stream of data between applications.

Key Characteristics of TCP

- **Connection-Oriented:** Establishes a connection before application data is exchanged.
- **Reliable Delivery:** Uses acknowledgements, sequence information, retransmission, and other mechanisms to support reliable delivery.
- **Ordered Delivery:** Presents data to the receiving application in the correct order.
- **Error Detection:** Uses checksums to detect corruption in transmitted TCP segments.
- **Retransmission:** Lost or unacknowledged data can be retransmitted.
- **Flow Control:** Regulates the amount of data sent so the receiver is not overwhelmed.
- **Port Numbers:** Identify the applications or services involved in communication.

TCP Three-Way Handshake

- **1. SYN:** The client requests connection establishment.
- **2. SYN-ACK:** The server acknowledges the request and sends its synchronization information.
- **3. ACK:** The client acknowledges the server, completing the handshake.

Common TCP Use Cases

- Web communication where TCP is used by protocols such as HTTP/1.1 and HTTP/2.
- File transfers where complete and ordered data is important.
- Email protocols such as SMTP, IMAP, and POP3.
- Other applications requiring reliable, ordered transport.

2. UDP — User Datagram Protocol

UDP is a **connectionless** transport protocol designed for simple, low-overhead datagram delivery. It does not provide TCP-style guarantees for delivery, ordering, or retransmission.

Key Characteristics of UDP

- **Connectionless:** Sends datagrams without establishing a connection first.
 - **No Delivery Guarantee:** UDP does not guarantee that a datagram will reach the destination.
 - **No Ordered Delivery:** Datagrams may arrive out of order.
 - **No Retransmission:** UDP itself does not automatically resend lost datagrams.
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- **Lightweight:** Uses fewer transport-level mechanisms than TCP, reducing protocol overhead.
- **Low Latency:** Its minimal transport mechanisms can suit time-sensitive applications.
- **Port Numbers:** Uses source and destination ports to identify applications.

Common UDP Use Cases

- DNS queries and other request/response traffic where low overhead is useful.
- Live media and streaming applications where timely data can be more important than retransmitting every lost packet.
- Online gaming and real-time applications.
- VoIP and other latency-sensitive communications.

3. TCP vs UDP

Feature	TCP	UDP
Full name	Transmission Control Protocol	User Datagram Protocol
Connection	Connection-oriented	Connectionless
Reliability	Provides reliability mechanisms	No built-in delivery guarantee
Ordering	Ordered byte stream	No ordering guarantee
Retransmission	Supported	Not provided by UDP itself
Flow control	Supported	Not provided by UDP itself
Overhead	Higher	Lower
Communication unit	Byte stream carried in TCP segments	Datagrams
Typical focus	Reliability and ordered delivery	Low overhead and timely delivery

4. How TCP and UDP Fit into Networking

Both TCP and UDP operate at the **Transport Layer**. They sit above the Internet Layer, which provides IP addressing and routing, and below the Application Layer, where network applications use transport services.

- **Application:** Creates or consumes application data.
- **Transport:** TCP or UDP adds transport information, including source and destination ports.
- **Internet:** IP provides logical addressing and routing information.
- **Network Access:** The packet is transmitted over the local network technology.

5. Choosing Between TCP and UDP

- **TCP:** Suitable when reliable, ordered, and complete delivery is important.
- **UDP:** Suitable when low overhead or timely delivery is important and the application can handle loss or reordering if needed.

Important: TCP is not simply “slow” and UDP is not simply “fast.” Actual performance depends on the application, network conditions, implementation, and protocol stack.

6. Cybersecurity Relevance

- TCP connections and flags can be examined during network traffic analysis.
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- Unexpected open TCP or UDP ports can help identify exposed services during security assessments.
- Security monitoring can analyze TCP sessions, UDP traffic, source/destination ports, and unusual communication patterns.
- Firewalls and network security controls commonly apply rules based on IP addresses, protocols, and ports.
- Understanding TCP and UDP helps analysts interpret packet captures and investigate network-based incidents.

Key Takeaway

TCP = connection-oriented, reliable, and ordered transport.

UDP = connectionless, lightweight, and without built-in guarantees for delivery or ordering.

Memory aid: TCP focuses on reliable delivery; UDP focuses on minimal transport overhead and timely delivery.